Cross-Battery State-of-Health Estimation under Randomized Loading: Leakage-Safe Evaluation, Uncertainty, and Model Interpretation

Thu Dang[1] and Thu Le[1]

FPT University, Ho Chi Minh City, Vietnam
minhthudang282@gmail.com, thulvm@fpt.edu.vn

Abstract. Estimating lithium-ion battery state of health (SOH) from randomized operating data becomes difficult when deployment involves a battery that was not observed during model development, because neighboring records from the same cell can otherwise make generalization appear stronger than it is. A leakage-safe pipeline is consequently constructed on the NASA PCoE RW1–RW28 benchmark, where target alignment and 50-step history engineering yield 548,010 operational observations from 28 cells and 14 model inputs. Cumulative Ah-throughput has a median within-cell Spearman coefficient of -0.9961 with SOH, whereas recent load-history variables retain only weak partial associations after cumulative exposure is controlled. Five regression families are compared across five reproducible cell-disjoint partitions containing 18 training, five calibration, and five test cells, and Random Forest obtains the lowest mean macro-cell MAE at 0.02445 ± 0.00410. Split-conformal intervals calibrated at a nominal 90% level reach mean macro-cell coverage of 0.9278, although EOL-specific diagnostics still reveal a missed-EOL rate of 0.2094. SHAP attribution and feature-family ablation point to the same dominant signal, because cumulative exposure receives the largest attribution and its removal increases macro-cell MAE by 98.29%. Together, these findings indicate that cross-battery SOH assessment is better characterized by whole-cell holdout, repeated comparison, uncertainty analysis, threshold-specific diagnostics, and transparent interpretation than by a single aggregate error score.

Keywords: Lithium-ion battery, state of health, randomized loading, cross-battery generalization, conformal prediction, model interpretation

1 Introduction

Lithium-ion battery state of health affects usable capacity, maintenance planning, and the operating margin available to energy-storage systems, while irregular demand makes those quantities harder to estimate from operational records alone. The challenge becomes sharper when the deployment target is a new battery

whose degradation path differs from the cells used during training, because row-wise train/test splits can place highly related observations from the same cell on both sides of the split and thereby inflate apparent accuracy. Reviews from both the SOH-estimation and prognostics communities continue to identify transfer across batteries and operating conditions as a central difficulty [2,20].

NASA's Randomized Battery Usage experiments are useful for examining that problem because irregular charge/discharge operation is periodically interrupted by reference procedures that anchor measured capacity [3]. The present analysis connects degradation characterization, repeated cross-battery model comparison, uncertainty assessment, and model interpretation within one evaluation pipeline, while complete cells are separated into training, calibration, and test roles before any final scoring is performed. Hyperparameter development is confined to training data, conformal residuals are derived only from calibration cells, and the held-out cells support the final accuracy, EOL, and interpretation diagnostics.

The scope remains deliberately narrow so that every result refers to one coherent experimental setting rather than mixing datasets with different protocols or target definitions. Within RW1–RW28, the central question is not whether a model can interpolate among familiar records, but whether the full analytical chain remains credible when the evaluated cell has never appeared during fitting or calibration.

2 Related Work

Battery SOH estimation spans electrochemical modeling, filtering, Gaussian-process regression, sparse Bayesian learning, tree ensembles, and neural networks [2,20]. Randomized and partial operating profiles are particularly relevant when controlled cycling is unavailable, so earlier work has adapted electrochemistry-based degradation modeling to randomized use [3], estimated capacity from short in-situ voltage segments with Gaussian-process regression [13], and explored data-driven degradation forecasting through relevance-vector and representation-learning frameworks [19,14,16]. The NASA battery repository also established a widely reused benchmark context for this line of work [17].

A second thread addresses transfer across batteries and operating domains, because strong performance on a familiar unit does not guarantee reliability on an unseen one. Transfer-learning and domain-generalization approaches have therefore been proposed for personalized SOH estimation and unknown operating environments [10,11,5,8,15], while conformal prediction provides a model-agnostic path from point estimates to empirical prediction intervals [18,1]. Table 1 summarizes how those representative references compare with the present pipeline across the capabilities emphasized in this paper.

Table 1. Capability comparison with representative battery SOH references. ✓ denotes explicit support and △ partial support.

Reference	Unseen unit	Random use	Unc.	EOL	XAI/ abl.	Repeated splits	Cost
Bole et al. [3]	–	✓	–	–	–	–	–
Richardson et al. [13]	△	–	✓	–	–	–	–
Yang et al. [16]	△	△	–	–	–	–	–
Chen et al. [5]	✓	–	–	–	–	–	–
Ma et al. [10]	✓	–	–	–	–	–	–
Mondal et al. [11]	✓	–	–	–	–	△	–
Our work	✓	✓	✓	✓	✓	✓	✓

The table positions the evaluation design used here: randomized-use data are tested on unseen cells, repeated partitions are retained, and uncertainty, EOL behavior, interpretation, and computational cost are reported within one pipeline. It does not imply that the cited methods are inferior outside these specific capabilities.

3 Methodology

3.1 Data and SOH Target Construction

The RW1–RW28 archive contains 1,117,894 records from 28 cells, among which an exact reference-discharge filter identifies 947 valid capacity measurements while charge and discharge rows form the operational exposure history. Each operational record is paired with the next valid reference-discharge capacity from the same cell, and SOH is normalized by that cell's initial valid reference capacity so that the target follows the next-reference logic implemented in the submitted notebook. Reference procedures themselves are excluded from cumulative operational exposure, while cell identity and target-derived fields are excluded from the predictor set. This construction makes the supervised problem operationally realistic, because the predictor side uses only information that would already be available at the moment of estimation, whereas the response comes from the next future reference check on the same cell. It also preserves the within-cell chronology that links irregular usage history to the next verified capacity state, which is essential when the later evaluation asks whether a model can transfer to a battery that was entirely absent from training and calibration.

Table 2. RW1–RW28 data and modeling representation from raw records to the final supervised table.

Item	Value
Cells	28
Raw records	1,117,894
Valid reference discharges	947
Operational charge/discharge rows	554,522
Rows with next-reference SOH	549,382
Rows with complete 50-step history	548,010
Engineered numeric predictors	12
Temperature-missingness indicators	2
Final model inputs	14
History window	50 steps
Operational EOL threshold	SOH ≤ 0.80

The reduction from raw records to the modeling table is driven mainly by target alignment: an operational row becomes usable only after a valid future reference capacity can be assigned from the same cell. The 50-step history requirement then removes only the short prefix of each cell, leaving 548,010 aligned observations for modeling.

In other words, the main loss of rows does not come from aggressive cleaning but from enforcing a consistent temporal learning setup. That choice is deliberate, because a shorter but leakage-safe table is more appropriate for the later unseen-cell evaluation than a larger table whose targets or predictors would rely on information unavailable at prediction time.

3.2 Feature Engineering and Cross-Battery Evaluation

Predictors are restricted to information available from the current and past operational history, with 12 numeric variables describing instantaneous electrical and thermal conditions, cumulative Ah-throughput, and 50-step rolling summaries. Two temperature-missingness indicators are added during modeling, which produces 14 final inputs after quality checks for missing targets, invalid history length, negative duration, negative Ah, and duplicate cell-step combinations have been completed.

The design therefore combines long-horizon exposure and short-horizon load descriptors instead of assuming that one representation alone is sufficient. Cumulative Ah-throughput captures broad degradation progression, whereas rolling summaries retain information about recent operating intensity and thermal context that may still refine the estimate once overall exposure has been accounted for.

Five model families are compared on the same GRU-valid test rows: Ridge, k-nearest neighbors (kNN), Random Forest, LightGBM, and a compact gated

recurrent unit (GRU) sequence model. Random Forest follows the bagged-tree formulation of Breiman [4], the classical baselines use scikit-learn [12], LightGBM follows the gradient-boosting framework of Ke et al. [7], and the recurrent model uses the GRU architecture introduced by Cho et al. [6]. For each of five reproducible seeds, the 28 cells are separated into 18 training cells, five calibration cells, and five test cells without overlap, while hyperparameters are developed with group-aware cross-validation on the seed-42 training cells and then frozen for the remaining partitions.

This repeated-partition design is intended to separate two questions that are often conflated in battery studies: whether a model performs well on average, and whether that conclusion remains stable when a different collection of complete cells is withheld. Reporting both absolute error and rank across the five partitions therefore provides a fuller picture of cross-battery robustness than a single split could offer.

3.3 Reliability and Model Interpretation

Nominal 90% split-conformal intervals are calibrated only from the five calibration cells in each partition, while pooled-row calibration serves as the primary procedure and a cell-balanced alternative gives every calibration cell equal residual weight. Test cells are used only to measure interval coverage and threshold behavior, with EOL defined as SOH ≤ 0.80 so that EOL-region error, missed-EOL rate, and pre-EOL false alarms can be separated from global regression metrics.

Predictive dependence is examined through two complementary mechanisms because attribution alone does not reveal how performance changes when information is removed. TreeSHAP [9] measures the magnitude of feature contributions across the held-out Random Forest evaluations, whereas feature-family ablation retrains and evaluates the model after coherent groups of inputs have been removed or restricted. Both procedures are interpreted as evidence about model dependence rather than as causal statements about electrochemical aging.

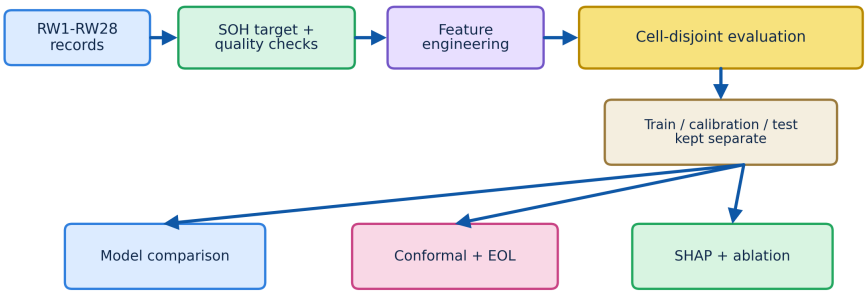


Fig. 1. RW1–RW28 analysis workflow from target construction to leakage-safe evaluation and final evidence.

Training, calibration, and test roles remain separated throughout the evaluation. Accuracy comparison, uncertainty diagnostics, and interpretation are produced only after the cell-disjoint split, so the final evidence does not feed back into model fitting.

4 Results

4.1 Degradation and Operational Exposure

RW10 is used first to make the target construction tangible, because SOH changes only when a new reference-capacity measurement becomes available while cumulative exposure continues to accumulate between those procedures.

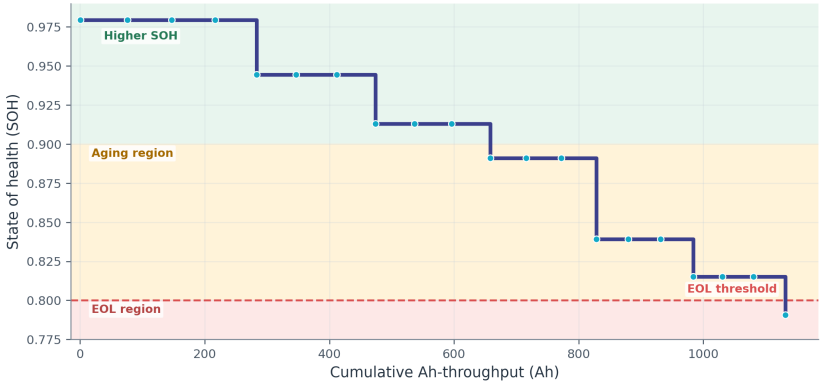


Fig. 2. RW10 SOH trajectory against cumulative Ah-throughput, with higher-SOH, aging, and EOL regions shown for reference.

The plateaus arise because many neighboring operational rows share the same next-reference SOH, whereas cumulative Ah-throughput continues to increase between reference procedures. This structure also motivates whole-cell holdout, since a row-wise split would distribute highly related observations across training and testing.

Across all 28 cells, the median within-cell Spearman association between cumulative Ah and SOH is -0.9961, whereas the recent-history partial correlations remain much smaller after exposure is controlled (-0.0656 for 50-step Ah, 0.1217 for 50-step mean current, and 0.0294 for 50-step maximum temperature). Figure 3 makes that between-cell heterogeneity visible without turning the association into a causal claim.

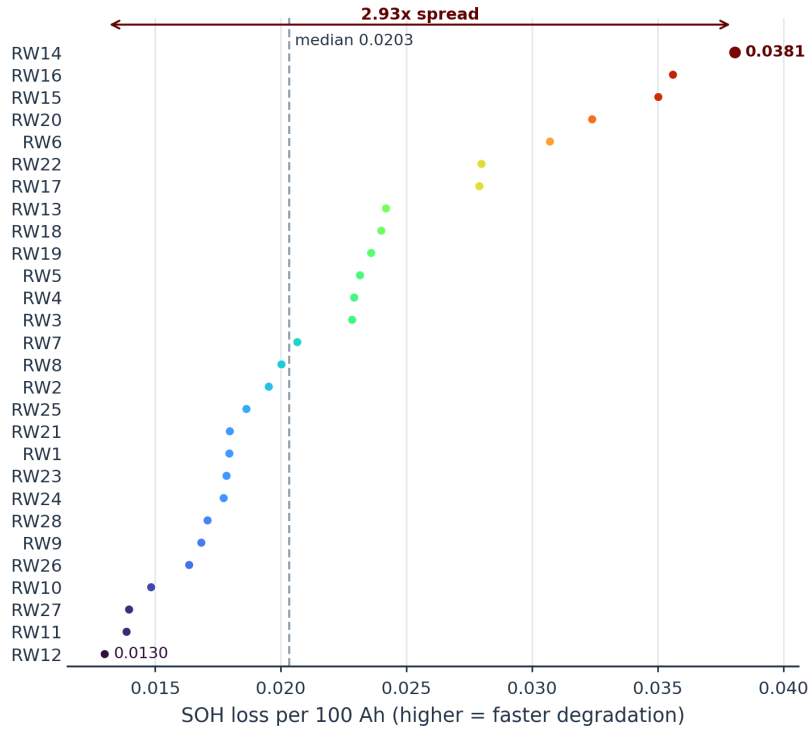


Fig. 3. Observed SOH-loss rate across the 28 RW cells; the dashed line marks the median of approximately 0.0203 SOH per 100 Ah.

The fastest cells lose roughly three times as much SOH per 100 Ah as the slowest cells, although every cell follows the same overall negative direction. The spread shows why a single aggregate regression score can conceal meaningful differences in degradation pace between batteries.

That heterogeneity also clarifies why the later comparison is summarized at the macro-cell level instead of only at the row level. If evaluation were dominated by batteries that contribute many rows or that degrade in a comparatively smooth manner, the resulting average error would understate the difficulty of generalizing to a broader set of unseen cells with different degradation speeds.

4.2 Generalization to Unseen Cells

The five candidate families are evaluated on identical unseen-cell rows, while macro-cell MAE gives every held-out battery equal weight.

This weighting scheme is important in the present setting because the held-out batteries do not contribute identical numbers of operational rows. A model that looks strong under a pooled-row average can still perform unevenly across cells, so the macro-cell view is used as the primary score and the row-level perspective is treated as supportive rather than definitive.

Table 3. Five-model comparison across cell-disjoint evaluation partitions; values are mean ± standard deviation over five reproducible splits.

Model	Macro-cell MAE	Macro-cell RMSE	Average rank
Random Forest	0.02445 ± 0.00410	0.03214 ± 0.00558	1.2
GRU	0.02643 ± 0.00481	0.03334 ± 0.00668	2.2
LightGBM	0.02772 ± 0.00199	0.03622 ± 0.00304	2.8
kNN	0.03362 ± 0.00993	0.04406 ± 0.01311	3.8
Ridge	0.04834 ± 0.00570	0.05924 ± 0.00548	5.0

Random Forest has the lowest mean macro-cell MAE, followed by GRU and LightGBM, while Ridge remains clearly separated from the nonlinear families. kNN improves on Ridge in mean error but has much larger across-split variability, indicating stronger sensitivity to the particular held-out cells.

The margin between Random Forest and GRU is modest in absolute terms, but the tree ensemble is more consistently near the top across the five partitions. That combination of lowest average error and comparatively stable ranking makes it the most defensible reference model for the subsequent uncertainty and interpretation analyses, even though no single split should be treated as a final verdict on model superiority.

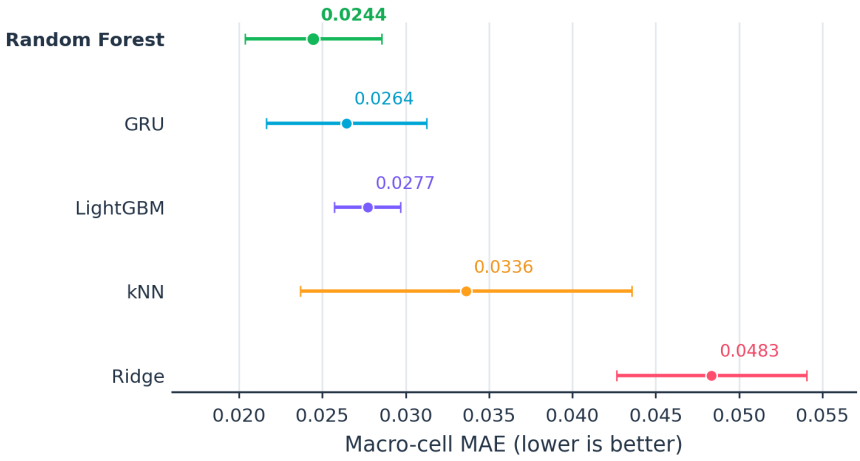


Fig. 4. Mean macro-cell MAE with across-split standard deviation for the five candidate models.

Random Forest combines the smallest mean error with moderate split-to-split variation, whereas kNN has the widest whisker and Ridge remains separated from the leading group. The figure intentionally avoids a shaded "best" zone because the evaluation protocol defines no such performance threshold.

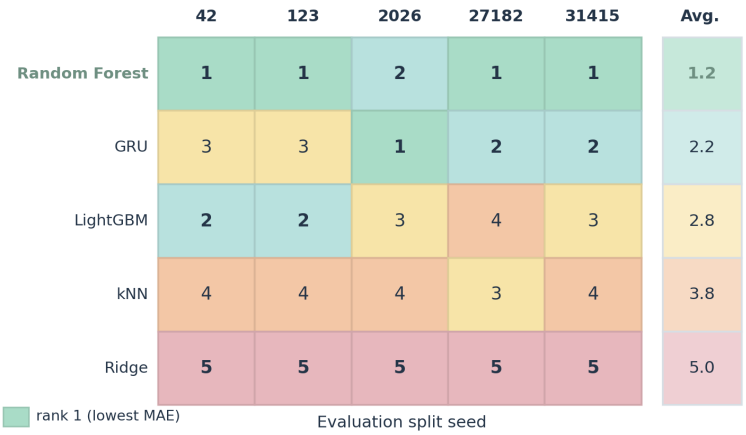


Fig. 5. Model rank across the five cell-disjoint partitions, where rank 1 denotes the lowest macro-cell MAE.

Seed 2026 reverses the top two models, while LightGBM moves between second and fourth place and Ridge remains fifth throughout. These changes show why the mean score is accompanied by repeated-split rank stability rather than being treated as universal model dominance.

Viewed together, Table 3, Fig. 4, and Fig. 5 suggest that the model-selection conclusion is directionally stable but not absolute. Random Forest is preferred because it performs best on average and remains competitive on every split, yet the seed-to-seed movements still justify the later emphasis on uncertainty and threshold-specific diagnostics rather than presenting one model as universally superior in every operating scenario.

4.3 Uncertainty and EOL Reliability

Threshold-specific point errors point in the same direction because the EOL-region macro-cell MAE reaches 0.04018 and the missed-EOL rate reaches 0.2094, while the pre-EOL false-alarm rate remains much lower at 0.0321. In practical terms, false negatives near retirement are more common than premature alarms under the point threshold used here, which is information that the global MAE cannot communicate on its own.

For that reason, interval reliability is reported alongside the point-threshold errors rather than being treated as a separate appendix-style result. A model can achieve an acceptable average error while still providing overly narrow uncertainty bands or missing a nontrivial share of near-EOL cases, so the coverage comparison below is intended to assess whether predictive uncertainty remains credible in the subset that matters most for retirement decisions.

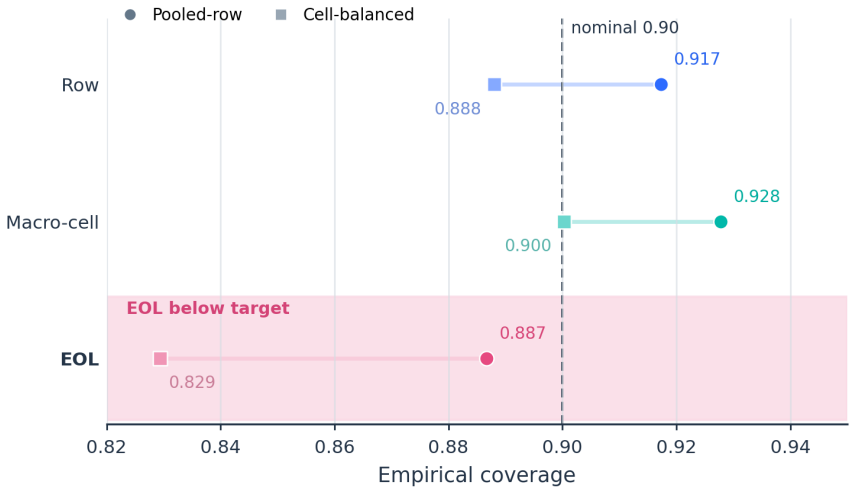


Fig. 6. Conformal coverage across row, macro-cell, and EOL views for pooled-row and cell-balanced calibration.

Pooled-row calibration exceeds the nominal 90% target at the row and macro-cell levels, whereas cell-balanced calibration is narrower on average but drops below nominal coverage at the row and EOL levels. The EOL difference, 0.8867 versus 0.8294, makes the reliability cost of the narrower intervals especially visible near the retirement threshold.

That trade-off is useful for interpretation because it separates interval sharpness from interval trustworthiness. In this dataset, the narrower cell-balanced intervals are not clearly preferable, since their reduced width is accompanied by visibly weaker protection in the EOL subset where under-coverage is the more consequential failure mode.

4.4 Model Interpretation

SHAP attribution assigns the largest mean absolute contribution to cumulative Ah at 0.07657, followed by 50-step Ah at 0.02756, 50-step maximum temperature at 0.02558, and instantaneous maximum temperature at 0.01832. Because attribution magnitude can be affected by correlated inputs, the ablation analysis in Fig. 7 is used as a second line of evidence rather than treating the SHAP ranking as sufficient by itself.

The interpretation section is therefore framed as a consistency check on what the winning model appears to rely on, not as an attempt to infer battery physics directly from a predictive algorithm. Agreement between attribution and ablation is taken as stronger evidence than either diagnostic in isolation.

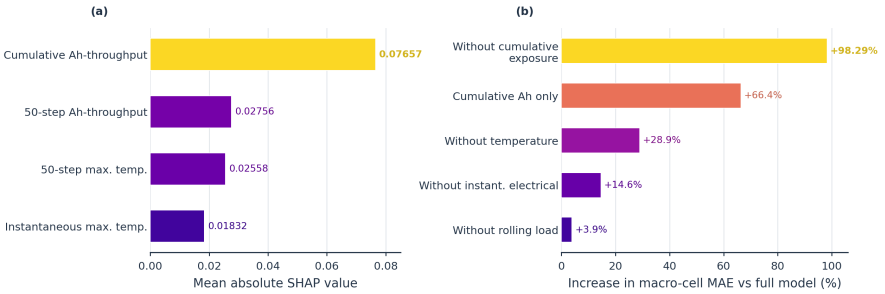


Fig. 7. Model interpretation views from SHAP and ablation.

Removing rolling-load features increases MAE by about 3.9%, whereas removing cumulative exposure raises it by 98.29%. Using cumulative Ah alone is also substantially worse than the full model, so long-term exposure is the dominant information family but does not fully replace the remaining inputs.

5 Discussion

Whole-cell holdout and repeated partitions remain necessary because degradation pace varies substantially across batteries even when cumulative Ah-throughput provides a stable exposure coordinate, while the leading models exchange rank across partitions. Reliability and interpretation add a second layer of evidence: coverage weakens in the EOL subset, and SHAP plus ablation both emphasize cumulative exposure without establishing a causal electrochemical mechanism.

Several limitations bound the conclusions because the records come from controlled NASA experiments rather than operational telemetry, the five partitions are reproducible stress scenarios rather than random draws from a deployment population, and the same held-out partitions later support the reliability characterization. External battery cohorts, broader loading environments, and prospective evaluation would be needed before the numerical performance reported here could be generalized beyond the RW1–RW28 setting.

6 Conclusion

The RW1–RW28 analysis supports a leakage-safe evaluation pipeline in which whole-cell holdout, repeated partitions, uncertainty, threshold-aware diagnostics, and interpretation are considered together. Cumulative Ah-throughput remains strongly associated with SOH decline, Random Forest achieves the lowest mean macro-cell MAE at 0.02445 ± 0.00410, pooled-row conformal calibration reaches 0.9278 mean macro-cell coverage, and both SHAP and ablation emphasize cumulative exposure; however, weaker EOL behavior shows why global error alone is not sufficient for cross-battery SOH assessment.

References

1. Angelopoulos, A.N., Bates, S.: Conformal Prediction: A Gentle Introduction. Foundations and Trends in Machine Learning 16(4), 494–591 (2023).
2. Berecibar, M., et al.: Critical Review of State of Health Estimation Methods of Li-ion Batteries for Real Applications. Renewable and Sustainable Energy Reviews 56, 572–587 (2016).
3. Bole, B., Kulkarni, C.S., Daigle, M.: Adaptation of an Electrochemistry-based Li-Ion Battery Model to Account for Deterioration Observed Under Randomized Use. Annual Conference of the Prognostics and Health Management Society 6 (2014).
4. Breiman, L.: Random Forests. Machine Learning 45(1), 5–32 (2001).
5. Chen, L., et al.: Domain Generalization-based State-of-Health Estimation of Lithium-ion Batteries. Journal of Power Sources 610, 234696 (2024).
6. Cho, K., et al.: Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation. In: Proceedings of EMNLP 2014, pp. 1724–1734 (2014).
7. Ke, G., et al.: LightGBM: A Highly Efficient Gradient Boosting Decision Tree. Advances in Neural Information Processing Systems 30, 3146–3154 (2017).
8. Lu, J., et al.: Deep Learning to Estimate Lithium-ion Battery State of Health without Additional Degradation Experiments. Nature Communications 14, 2760 (2023).
9. Lundberg, S.M., Lee, S.-I.: A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems 30, 4765–4774 (2017).
10. Ma, G., et al.: A Transfer Learning-Based Method for Personalized State of Health Estimation of Lithium-Ion Batteries. IEEE Transactions on Neural Networks and Learning Systems 35(1), 759–769 (2024).
11. Mondal, A., et al.: State-of-Health Estimation of Li-Ion Batteries Using Semi-parametric Adaptive Transfer Learning. IEEE Transactions on Transportation Electrification 10(1), 1080–1088 (2024).
12. Pedregosa, F., et al.: Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research 12, 2825–2830 (2011).
13. Richardson, R.R., et al.: Gaussian Process Regression for In Situ Capacity Estimation of Lithium-Ion Batteries. IEEE Transactions on Industrial Informatics 15(1), 127–138 (2019).
14. Severson, K.A., et al.: Data-driven Prediction of Battery Cycle Life before Capacity Degradation. Nature Energy 4, 383–391 (2019).
15. Tan, R., et al.: Forecasting Battery Degradation Trajectory under Domain Shift with Domain Generalization. Energy Storage Materials 72, 103725 (2024).
16. Yang, N., et al.: Robust State of Health Estimation of Lithium-Ion Batteries Using Convolutional Neural Network and Random Forest. Journal of Energy Storage 48, 103857 (2022).
17. Saha, B., Goebel, K.: Battery Data Set. NASA Ames Prognostics Data Repository, Moffett Field, CA (2007).
18. Vovk, V., Gammerman, A., Shafer, G.: Algorithmic Learning in a Random World. Springer, New York (2005).
19. Wang, D., et al.: Prognostics of Lithium-Ion Batteries Based on Relevance Vectors and a Conditional Three-Parameter Capacity Degradation Model. Journal of Power Sources 239, 253–264 (2013).
20. Zhang, J., Lee, J.: A Review on Prognostics and Health Monitoring of Li-Ion Battery. Journal of Power Sources 196(15), 6007–6014 (2011).